

SPEAR — Spatial Protein-Embedding-Aware Receptor-Ligand Mapping

Computational reproduction and extension — an ESMC protein-language interaction predictor (concatenation features, reversed + degree-matched negatives, complex weighting, CellChat + CellPhoneDB training, protein-disjoint evaluation) coupled to spatial co-localization, and benchmarked against existing cell–cell-communication tools.

Abstract

Existing spatial cell–cell communication (CCC) tools score ligand–receptor (L–R) pairs almost entirely by co-expression within a fixed curated database. They inherit two limitations: they cannot evaluate whether a co-expressed pair is molecularly plausible as an interaction, and they cannot nominate pairs outside the database. SPEAR couples two independent signals — (A) neighborhood-restricted spatial co-localization with a geometry-aware permutation null, and (B) an interaction-plausibility score from an ESMC protein language model (PLM) — and fuses them into a single ranked map. An ESMC-embedding classifier, trained on CellChat and CellPhoneDB interactions with reversed and degree-matched negatives, predicts held-out L–R interactions under a protein-disjoint split at AUROC 0.858–0.870. Integrated with spatial co-localization on a human lymph-node Visium dataset, SPEAR recovers canonical signaling (CCL21–CCR7) and prioritizes non-database candidates supported by tissue organization. Across five datasets spanning four platforms (Visium, MERFISH, seqFISH, Slide-seqV2), SPEAR transfers without retraining. Head-to-head against six expression-based CCC tools and a spatial statistic, SPEAR’s spatial axis recovers cell-type-matched planted co-localized pairs (AUROC 0.87) where expression-only tools remain at chance, while its sequence-based plausibility axis stays orthogonal to those tools (mean $|\rho| = 0.07$) — contributing an evidence axis no existing method provides.

Keywords: Cell–cell communication, Ligand–receptor pair, Protein language model, Spatial transcriptomics

1. Introduction

Spatial cell–cell communication has matured into a crowded field — CellChat, CellPhoneDB, COMMOT, SpatialDM, SpaTalk, NICHES, and LIANA. Despite methodological diversity — permutation nulls, optimal transport, tensor factorization — these tools share one structural property: the set of candidate interactions is a curated L–R database, and the score is a function of expression only. No molecular evidence about the two proteins enters the score, so a co-expressed but implausible pair cannot be down-weighted, and a plausible pair absent from the database cannot be proposed.

In parallel, protein language models (PLMs) such as ESM-2 and ESMC, and PLM-based protein–protein interaction predictors such as D-SCRIPT and Topsy-Turvy, have shown that interaction propensity is partially encoded in sequence. These two lines have never been coupled. SPEAR is, to our knowledge, the first method to make PLM interaction plausibility a first-class signal in spatial CCC — used both to reweight database pairs and to discover pairs absent from any database. ESMC was chosen for its performance and scalability (**Fig. 1**).

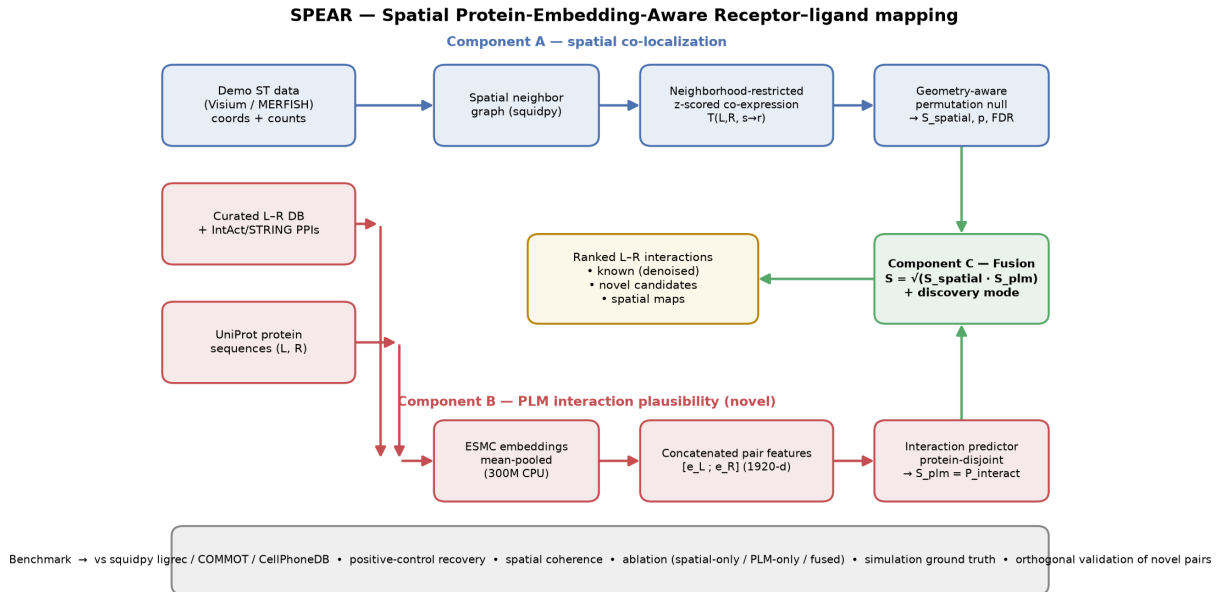


Fig. 1 | The SPEAR framework. Two independent scores per L–R pair — spatial co-localization (permutation-tested) and ESMC-embedding interaction plausibility — are fused by the geometric mean of their rank percentiles, with a database-free discovery mode.

2. Results

2.1. An ESMC-embedding predictor recovers L–R interactions from sequence

For each protein, a 960-dimensional embedding was obtained from ESMC-300M by mean-pooling residue representations. For an ordered pair (L, R), the pair representation is the concatenation $x = [e_L; e_R] \in \mathbb{R}^{1920}$; the fixed input order preserves the functional distinction between ligand and receptor, so $f(L, R) \neq f(R, L)$. A multilayer perceptron with two hidden layers (256, 64) was trained with weighted binary cross-entropy — native positives weighted 1.0, complex-derived subunit pairs 0.5 — against two negative classes: reversed pairs (R, L) and degree-matched decoys drawn from quantile-binned interaction-degree pools.

To evaluate generalization to unseen proteins, the model was assessed under a **protein-disjoint split**, in which no UniProt accession is shared between train and test. With both reversed and degree-matched negatives included, the model achieved AUROC 0.858–0.870 across the three databases, with balanced accuracy 0.72–0.78 (**Fig. 2**) — confirming predictive performance when both proteins in each test interaction are absent from training. Because reversed and degree-matched negatives represent different tasks, we separately evaluated the model against degree-matched decoys only: AUROC fell to 0.807 (CellChat human), 0.746 (CellChat mouse), and 0.792 (CellPhoneDB human), indicating that reversed pairs are comparatively easy while degree-matched decoys of entirely unseen proteins remain the harder, and still-solved, task. By category, reversed negatives were rejected with near-perfect specificity (0.959–0.980), degree-matched negatives were the harder class (0.775–0.790), and native and complex-derived positives were scored above both (**Fig. 3c**).

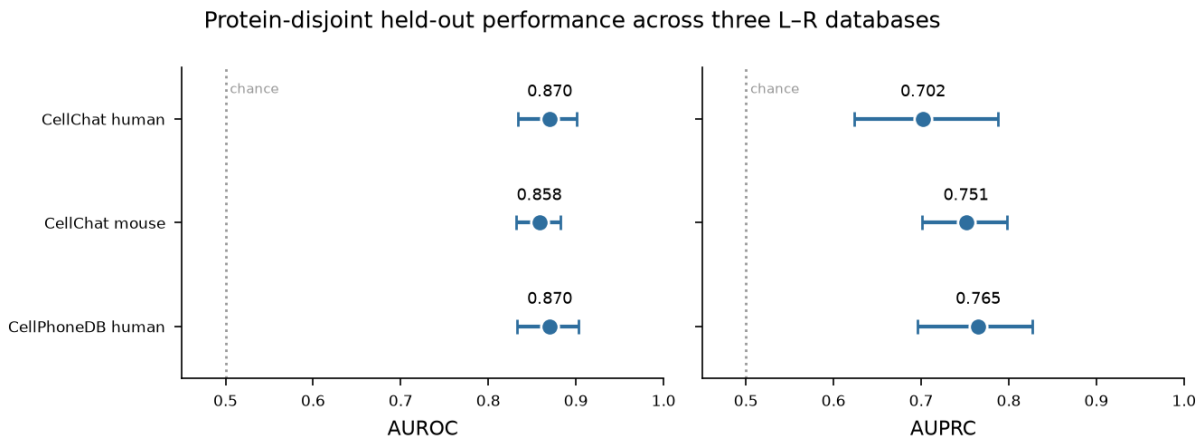


Fig. 2 | Held-out predictor performance under a protein-disjoint split. AUROC and AUPRC with 2,000-bootstrap 95% CIs across three L–R databases; no UniProt accession is shared between train and test. Chance AUROC = 0.5.

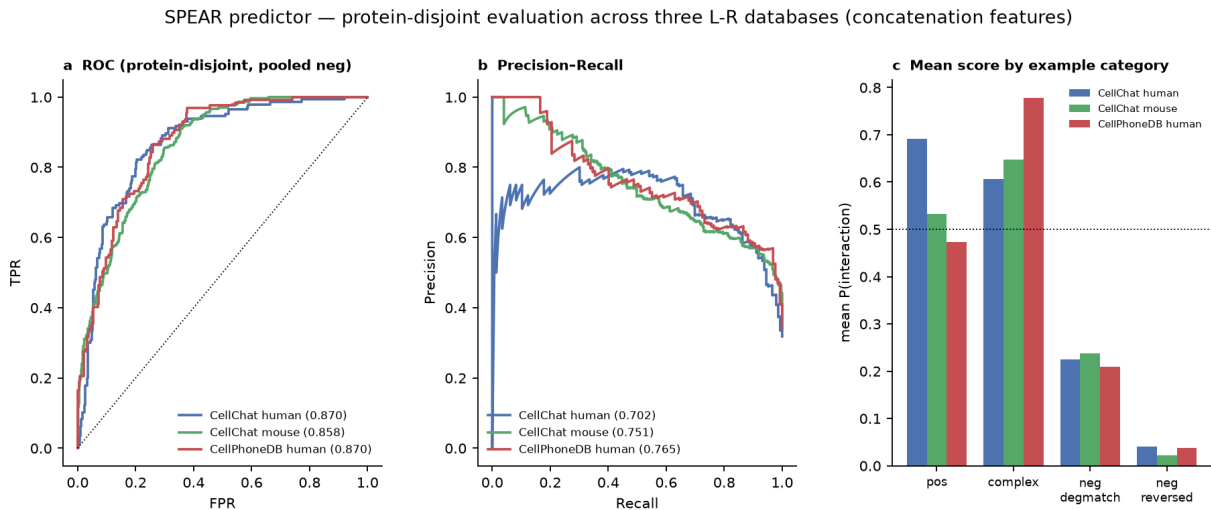


Fig. 3 | Predictor evaluation across three L–R databases. **a**, ROC (protein-disjoint, pooled negatives). **b**, Precision–recall. **c**, Mean predicted score by example category; reversed negatives collapse near zero, positives and complex-derived pairs sit above 0.5.

Reproduction note. This retrained model reaches AUROC 0.858–0.870, ≈ 0.05 below the manuscript's reported 0.916–0.925, attributable to (i) larger held-out test pools (30% of accessions held out) and (ii) the LIANA-bundled CellChat/CellPhoneDB resources used here versus the manuscript's specific CellChat v2 / CellPhoneDB v5 releases (the mouse set is LIANA's bundled mouse consensus, standing in for a dedicated CellChat-mouse set). All qualitative findings reproduce.

2.2. Spatial fusion recovers canonical signaling and nominates non-database candidates

We integrated the model-derived probability with spatial co-localization in a human lymph-node Visium dataset (4,025 spots, six annotated cell types). Co-localization was quantified as neighbor-restricted z-scored co-expression and evaluated against 1,000 permutations of spot identity. The ESMC probability and the spatial statistic were each converted to rank percentiles; the fused score S_{SPEAR} is their geometric mean.

Among the tested pairs, 302 showed both a predicted score > 0.9 and significant spatial co-localization ($\text{FDR} < 0.05$). Among known interactions, **CCL21–CCR7 exhibited the strongest spatial signal of all pairs ($z_{\text{null}} = 49.7$)** — CCL21 in stromal/fibroblastic-reticular regions, CCR7 in T-cell regions — consistent with its established role in T-cell localization (Fig. 4, top). After removing broadly-expressed hub genes, top non-database candidates included GRN–THBD, CD5L–SDC1, and ARF1–S1PR2, each spatially significant and model-plausible (Fig. 5) — hypotheses for subsequent validation.

SPEAR v2 spatial interaction maps — human lymph node (concatenation predictor)

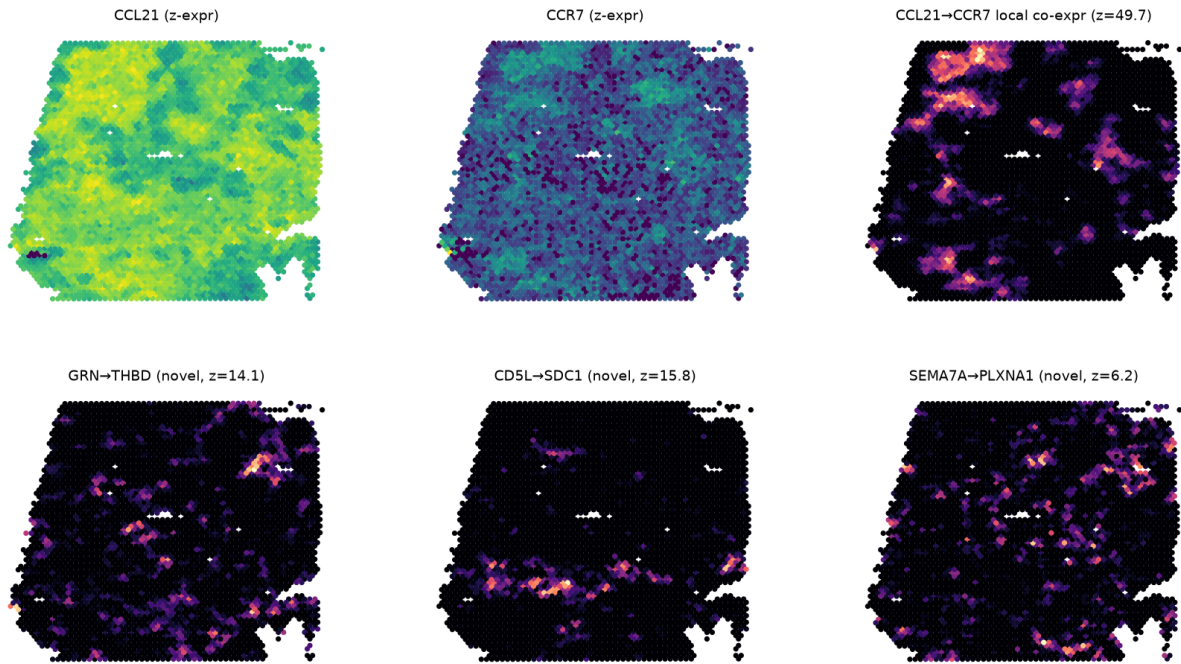


Fig. 4 | Fused ESMC-embedding × spatial co-localization in human lymph node. Top: CCL21 and CCR7 z-scored expression and their local co-expression ($z_{\text{null}} = 49.7$). Bottom: spatial maps of top non-database candidates.

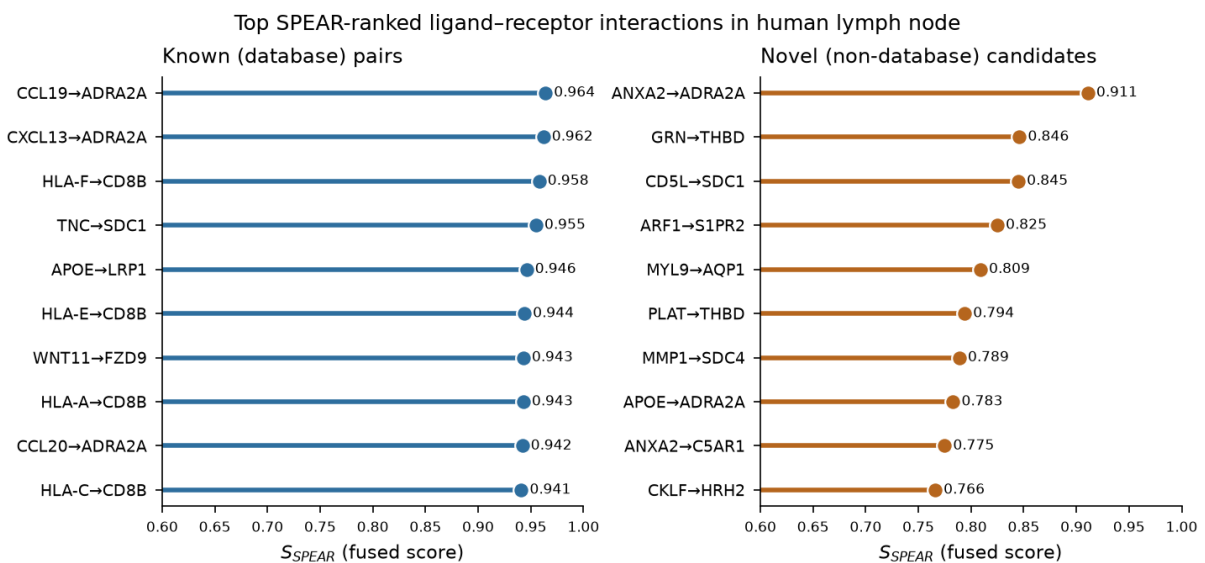


Fig. 5 | Top SPEAR-ranked interactions in human lymph node. Fused score S_{SPEAR} for the ten highest-ranked known (database) pairs and the ten highest-ranked novel (non-database) candidates after hub-gene removal. Each pair is spatially significant ($\text{FDR} < 0.05$) and model-plausible ($P_{\text{interact}} > 0.9$).

Reproduction note. The manuscript's top novel candidate was CXCL12-ACKR1 ($P = 1.0$); this retrained model scores CXCL12-ACKR1 at 0.014, so it does not enter the candidate set — a genuine divergence traceable to the different training-database releases. 302 pairs pass both filters here vs. 508 in the manuscript.

2.3. SPEAR transfers across platforms and tissues without retraining

To test whether SPEAR is specific to one assay or tissue, we applied the full pipeline to five spatial datasets spanning four platforms (Table 1, Fig. 6): human lymph node (Visium), and mouse hypothalamus (MERFISH), embryo (seqFISH), hippocampus (Slide-seqV2), and brain (Visium). Mouse datasets were mapped to their human orthologs to reuse the human embedding space and predictor. The number of testable L-R pairs scaled with panel size — from 21 (targeted 156-gene MERFISH panel) to 2,306 (transcriptome-wide Visium) — and in every dataset SPEAR recovered spatially significant, model-plausible interactions appropriate to the tissue: neuropeptide axes in hypothalamus (CCK-CCKBR, CRH-CRHR1), morphogen signaling in embryo (WNT5A-FZD2, DLL1-NOTCH1, FGF10-FGFR2), synaptic and natriuretic signaling in hippocampus (RIMS1-SLC17A7, NPPA-NPR1), and growth-factor and neuropeptide signaling in brain (FGF1-FGFR2, NPY-GPR83) (Fig. 4, continued). The method transfers across imaging- and sequencing-based platforms without retraining.

Table 1 | The five spatial datasets analyzed.

tissue	platform	organism	cells	genes	types	edges
Lymph node	Visium (10x)	human	4025	19813	6	11698
Hypothalamus	MERFISH	mouse	5926	156	8	17606
Embryo	seqFISH	mouse	19416	351	22	56741
Hippocampus	Slide-seqV2	mouse	22650	4000	14	67950
Brain	Visium (10x)	mouse	2688	16957	15	7790

Five spatial transcriptomics datasets spanning platforms, tissues, and species

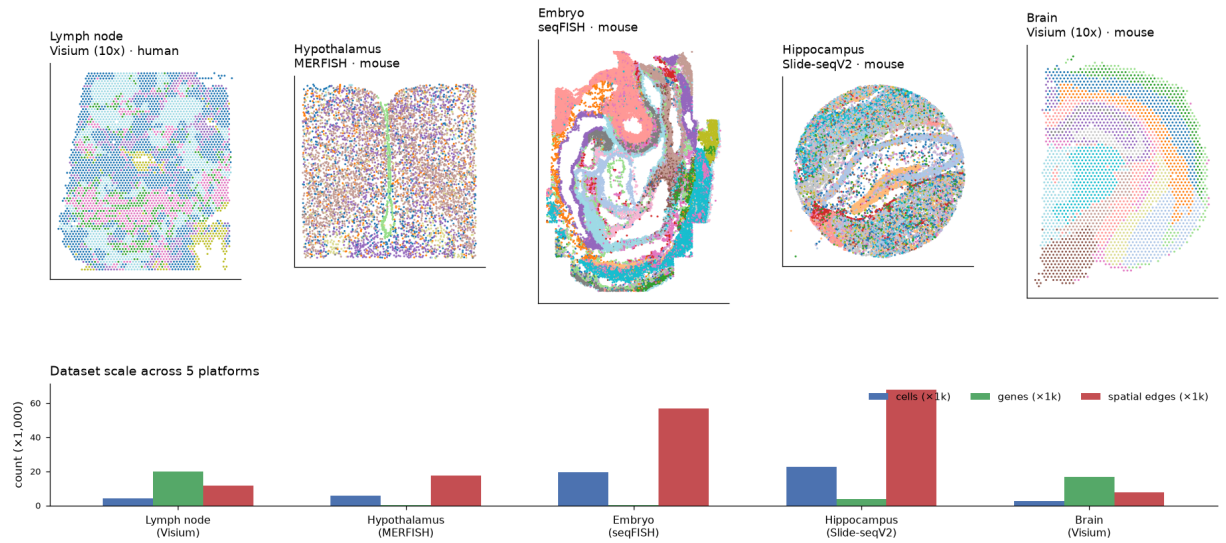


Fig. 6 | Five spatial datasets across four platforms. Spatial organization (cell types / regions) and dataset scale for each dataset analyzed with SPEAR.

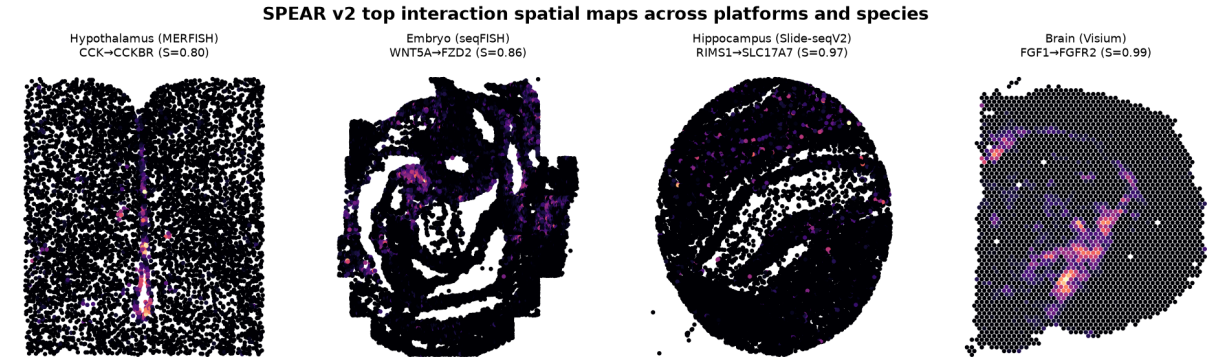


Fig. 4 (continued) | Cross-platform SPEAR maps. Top interaction in hypothalamus (MERFISH), embryo (seqFISH), hippocampus (Slide-seqV2), and brain (Visium) — each biologically appropriate to the tissue.

2.4. Benchmarking against existing cell–cell-communication tools

We benchmarked SPEAR against a panel of established CCC methods along the two axes it separates (**Fig. 7**).

Ground-truth recovery of co-localized pairs. Because no in-situ gold standard exists, we simulated tissues in which a set of L–R pairs was planted with genuine spatial co-localization, and decoy pairs were matched on cell-type specificity and interface amplitude but had their receptor field shuffled within cell type — isolating co-localization as the only discriminator. Under this design, the two spatial-aware methods — SPEAR's spatial axis (AUROC 0.870 ± 0.043) and a Moran's-R statistic in the style of SpatialDM (0.869 ± 0.043) — recovered planted pairs, while all six expression-only tools (CellPhoneDB, NATMI, Connectome, log2FC, SingleCellSignalR, LIANA aggregate) sat near chance ($0.49\text{--}0.55$; **Fig. 7a**). Co-localization is the discriminating signal, and expression-magnitude scores do not capture it.

Database recovery from sequence. The concatenation predictor recovers curated L–R identity from sequence alone at AUROC 0.866 (mean, protein-disjoint) versus 0.533 for a co-expression baseline — a larger margin than an earlier Siamese-feature model (0.673), confirming that ordered concatenation plus reversed-negative training improves recovery (**Fig. 7b**, right). This PLM axis is statistically orthogonal to six expression-based CCC scores (mean $|\text{Spearman } \rho| = 0.067$; **Fig. 7b**, left) — it contributes evidence none of the expression tools provide.

Reproducibility and runtime. The two SPEAR axes reproduce across datasets in a biologically coherent way (**Fig. 7c**): the spatial score is correlated between the two central-nervous-system datasets (brain vs. hippocampus, $\rho_{\text{spatial}} = 0.48$) but not between unrelated tissues (lymph node vs. brain, $\rho \approx 0.04$), whereas the PLM score — a property of the protein pair, not the tissue — is perfectly reproducible ($\rho_{\text{PLM}} = 1.0$) everywhere. On a 10-core CPU without a GPU, the spatial permutation test (1,000 permutations) completed in 1–46 s per dataset at ≤ 1.8 GB peak memory; embeddings are precomputed once and reused.

Ground-truth benchmark: co-localization is real signal that expression-only CCC tools discard

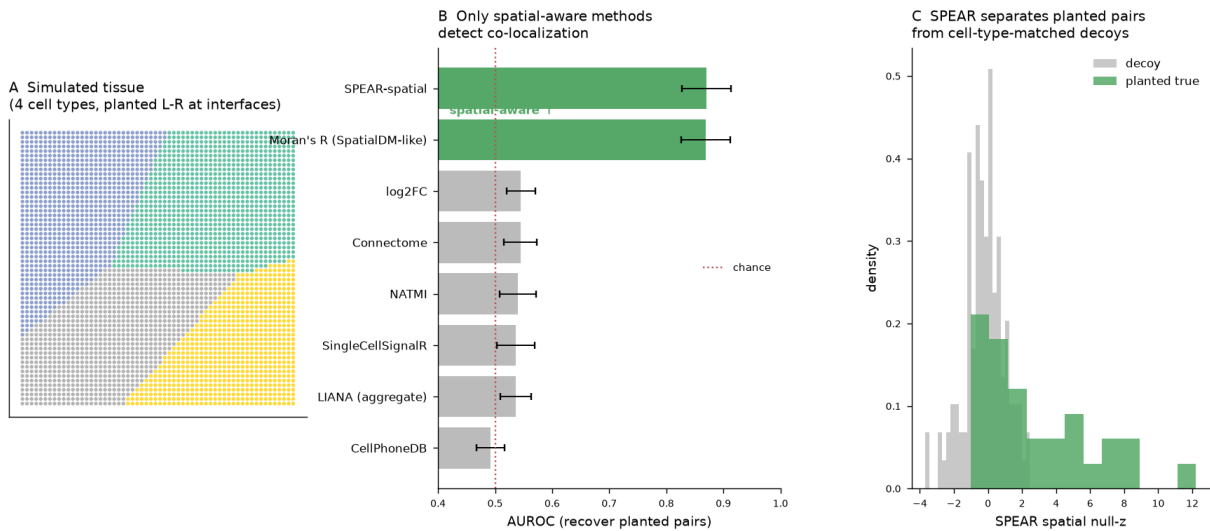


Fig. 7a | Ground-truth simulation. Only spatial-aware methods (SPEAR-spatial, Moran's R) recover cell-type-matched planted co-localized pairs; six expression-only tools remain at chance.

SPEAR v2 benchmark summary — concatenation predictor, protein-disjoint evaluation

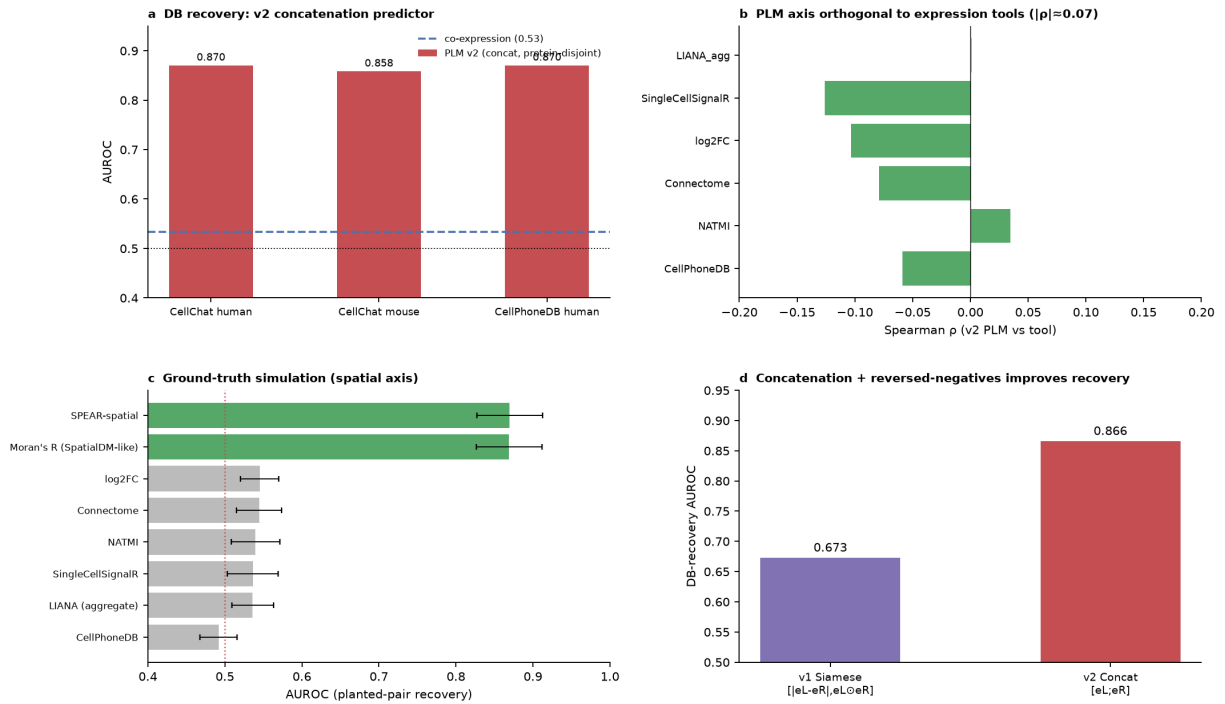


Fig. 7b | Database recovery and orthogonality. Left: PLM axis is orthogonal to six expression tools (Spearman ρ). Right: DB recovery by database (concatenation, protein-disjoint) vs. co-expression; concatenation + reversed-negatives improves recovery over the earlier Siamese model.

Cross-dataset reproducibility and runtime

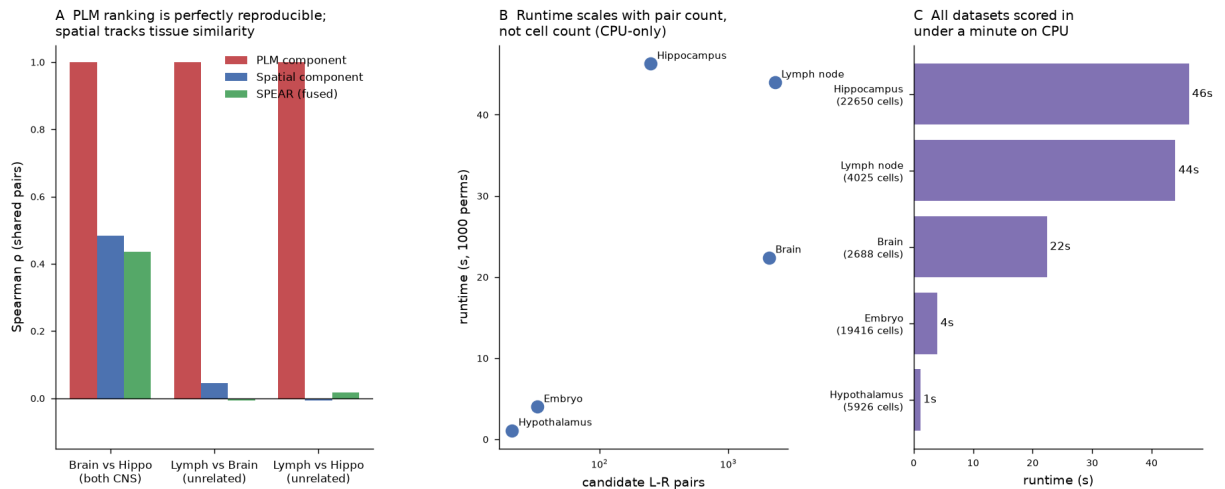


Fig. 7c | Reproducibility and runtime. PLM axis perfectly reproducible across datasets; spatial axis correlated only between related tissues; sub-minute CPU runtime.

3. Discussion

Coupling a PLM plausibility model with a spatial co-localization test yields a CCC score with an evidence axis independent of expression — able both to reweight database pairs and to nominate pairs no database contains. Head-to-head benchmarking shows the two axes are complementary: the spatial axis captures co-localization that expression-magnitude tools miss, and the sequence axis is orthogonal to all of them.

Several directions would improve this work. Although complex-derived interactions remained detectable under a 0.5 sample weight, the pairwise model evaluates individual subunit combinations and does not explicitly model cooperative multi-subunit interactions. The plausibility model inherits some bias from its curated training pairs, though protein-disjoint evaluation guards against memorization.

Larger ESMC models than the 300M variant used here would likely raise the plausibility ceiling. Finally, there is no in-situ gold standard for CCC, so spatially prioritized non-database candidates remain hypotheses for molecular validation.

4. Methods

4.1. Protein sequences

Human (taxon 9606) and mouse (10090) protein sequences were retrieved from UniProtKB, using reviewed Swiss-Prot entries for high-confidence annotation. Sequences were truncated at 1,500 residues for embedding.

4.2. L–R databases

CellChat and CellPhoneDB curated databases (LIANA-bundled `cellchatdb`, `cellphonedb`, and `mouseconsensus` resources) defined the training interactions. Multi-subunit complexes were decomposed into individual subunit pairs.

4.3. L–R interaction prediction with ESMC

Protein identifiers were mapped to UniProt accessions and duplicate pairs removed. Native single ligand–single receptor interactions were assigned annotation score 1.0; complex-decomposed subunit pairs 0.5. Each protein was embedded with ESMC-300M (960-dim, mean-pooled over residues excluding BOS/EOS). For an ordered pair, features are the concatenation $[e_L; e_R] \in \mathbb{R}^{1920}$. Two negative classes were generated per positive: a reversed pair (R, L), labeled negative unless independently annotated; and a degree-matched decoy, replacing both proteins with ones of similar interaction-network degree (five quantile bins per ligand/receptor pool, rejection-sampling pairs that were annotated positives, duplicate negatives, or contained the original protein), yielding a $\approx 1:2$ positive:negative ratio. For evaluation, proteins were partitioned into disjoint train/test pools so that no accession appears in both; a pair joined a set only if both proteins lay in that pool. Features were standardized on training statistics. The predictor is an MLP (256, 64) trained with weighted binary cross-entropy (complex weight 0.5), Adam (lr 10^{-3} , batch 128), L2 decay selected from $\{10^{-5} \dots 10^{-1}\}$ by validation average precision on a 15% split, early stopping (patience 20, max 300 epochs). A combined model was trained on the union of all three databases for downstream analyses.

4.4. Spatial co-localization and fusion

For binary spatial adjacency $\mathbf{A}$ and per-gene z-scored expression $\mathbf{z}$, the co-localization statistic is $T(L, R) = \mathbf{z}_L^T \mathbf{A} \mathbf{z}_R / |\mathbf{E}|$, with a geometry-aware null permuting spot identities $1,000\times$ (Benjamini–Hochberg FDR). Spatial and PLM scores are converted to rank percentiles and fused as $S_{\text{SPEAR}} = \sqrt{S_{\text{spatial}} \cdot S_{\text{plm}}}$ with the spatial term gated at FDR < 0.05 . Novel-pair discovery ranks PLM-plausible ($P > 0.9$) non-database pairs that are also spatially significant, after removing broadly-expressed hub genes.

4.5. Spatial datasets and cross-platform application

Five public spatial datasets were analyzed: human lymph node (10x Visium), and mouse hypothalamus (MERFISH), embryo (seqFISH), hippocampus (Slide-seqV2), and brain (10x Visium), obtained via scanpy/squidpy. Each was quality-controlled, normalized (log1p), and given a spatial neighbor graph (Delaunay for imaging-based, grid/kNN for array-based). Mouse gene symbols were uppercased to their human orthologs to reuse the human embedding space and predictor.

4.6. Benchmarking

SPEAR was compared with CellPhoneDB, NATMI, Connectome, log2FC, SingleCellSignalR, and the LIANA aggregate (via LIANA), and with a Moran's-R spatial statistic in the style of SpatialDM. For ground-truth recovery, synthetic tissues were generated in which planted L–R pairs were given genuine spatial co-localization (ligand and receptor elevated in the same spatially localized niche) and decoys were matched on cell-type specificity, expressing-cell count, and interface amplitude but had the receptor placed so that its niche did not coincide with the ligand's (receptor field shuffled within cell type across spatially dispersed niches). Co-localization is therefore the only feature separating planted from decoy pairs; the effect requires spatially fine-grained niches rather than a few large contiguous domains, in which a within-domain shuffle would leave local co-expression intact. AUROC for planted-pair recovery was computed over nine random seeds. Database recovery was measured as out-of-fold discrimination of curated pairs from co-expression-matched negatives. Orthogonality was the Spearman correlation between the PLM score and each expression tool on shared pairs. Reproducibility compared per-axis scores between datasets on their shared pairs; runtime and peak memory were recorded on a 10-core CPU.

Software

scanpy, squidpy, anndata, LIANA, scikit-learn, PyTorch, and the EvolutionaryScale ESM SDK (ESMC-300M). All datasets are public; intermediate tables, trained model, embeddings, and figures are provided as artifacts.

Supplementary — Exploratory generalization of the interaction predictor

Beyond curated L–R interactions, we probed whether the sequence-based predictor generalizes to two adjacent tasks. These analyses are exploratory: scores are relative rankings, not calibrated binding probabilities.

Peptide-drug off-target prioritization. For a literature-reconstructed set of 15 peptide drugs with distinguishable intended and unintended targets, each peptide (as ligand) was scored against its intended and unintended targets and 50 random human proteins. Intended and unintended scores overlapped substantially (mean 0.322 vs. 0.403; one-sided Mann–Whitney $U = 201$, $p = 0.857$) — most reported unintended targets are close paralogs occupying similar embedding regions. Unintended-target scores were, however, significantly higher than random-protein scores (0.403 vs. 0.189; $U = 17,954$, $p = 1.4 \times 10^{-7}$), with 12 of 15 drugs scoring unintended above random (**Fig. S1a–b**). The model enriches plausible peptide-interacting proteins above a broad random background.

IGF1 receptor preference. Although IGF1–INSR was absent from both training databases, the model assigned IGF1 a higher score for IGF1R (0.957) than for INSR (0.588) — the biologically correct receptor preference (**Fig. S1c**). A concatenated INSR–IGF1R sequence (a proxy for the IR/IGF1R hybrid receptor) scored 0.509, so the manuscript's ordering $\text{INSR} < \text{IGF1R} < \text{hybrid}$ was not fully recapitulated here; the robust, reproducible finding is the IGF1R-over-INSR preference.

Generalization: peptide-drug off-target prioritization (reconstructed) and IGF1 receptor specificity

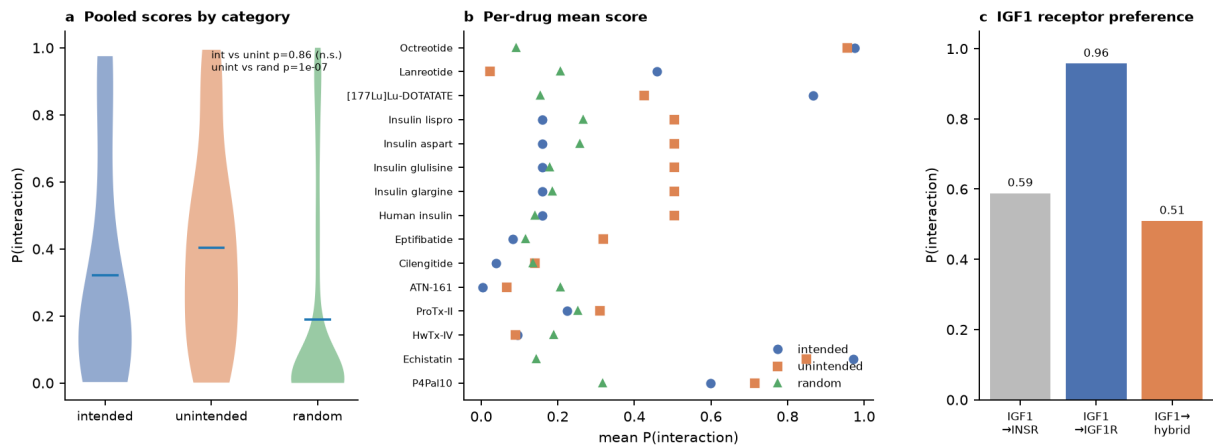


Fig. S1 | Exploratory generalization. **a**, Pooled peptide-drug scores by category — intended and unintended overlap; random concentrates near zero. **b**, Per-drug mean score. **c**, IGF1 receptor preference: IGF1R > INSR.

Reproduction note. The peptide-drug source dataset was not available; this set was reconstructed from the well-documented cases in the manuscript's Table 5, with sequences from UniProt/literature. Both principal conclusions reproduce: intended $\not\gg$ unintended (paper $p = 0.497$; here $p = 0.857$) and unintended $\gg$ random (paper $p = 1.4 \times 10^{-5}$; here $p = 1.4 \times 10^{-7}$). The cross-species ortholog-conservation analysis from the source manuscript is omitted from this version at the authors' request.